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Influence-matrix diagnostic of a data assimilation system

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SUMMARY

The influence matrix is used in ordinary least-squares applications for monitoring statistical multiple-regression analyses. Concepts related to the influence matrix provide diagnostics on the influence of individual data on the analysis—the analysis change that would occur by leaving one observation out, and the effective information content (degrees of freedom for signal) in any sub-set of the analysed data. In this paper, the corresponding concepts have been derived in the context of linear statistical data assimilation in numerical weather prediction. An approximate method to compute the diagonal elements of the influence matrix (the self-sensitivities) has been developed for a large-dimension variational data assimilation system (the four-dimensional variational system of the European Centre for Medium-Range Weather Forecasts). Results show that, in the boreal spring 2003 operational system, 15% of the global influence is due to the assimilated observations in any one analysis, and the complementary 85% is the influence of the prior (background) information, a short-range forecast containing information from earlier assimilated observations. About 25% of the observational information is currently provided by surface-based observing systems, and 75% by satellite systems.

Low-influence data points usually occur in data-rich areas, while high-influence data points are in data-sparse areas or in dynamically active regions. Background-error correlations also play an important role: high correlation diminishes the observation influence and amplifies the importance of the surrounding real and pseudo observations (prior information in observation space). Incorrect specifications of background and observation-error covariance matrices can be identified, interpreted and better understood by the use of influence-matrix diagnostics for the variety of observation types and observed variables used in the data assimilation system.

KEYWORDS: Observations Regression methods

1. INTRODUCTION

Over the years, data assimilation schemes have evolved into very complicated systems, such as the four-dimensional variational system (4D-Var; Rabier *et al.* 2000) at the European Centre for Medium-Range Weather Forecasts (ECMWF). The scheme handles a large variety of both space- and surface-based meteorological observations. It combines the observations with prior (or background) information of the atmospheric state and uses a comprehensive (linearized) forecast model to ensure that the observations are given a dynamically realistic, as well as statistically likely response in the analysis.

Effective monitoring of such a complex system, with the order of 10^7 degrees of freedom and more than 10^6 observations per 12 h assimilation cycle, is a necessity. The monitoring cannot be restricted to just a few indicators, but a complex set of measures is needed to indicate how different variables and regions influence the data assimilation (DA) scheme. Measures of the observational influence are useful for understanding the DA scheme itself. How large is the influence of the latest data on the analysis and how much influence is due to the background? How much would the analysis change if one single influential observation were removed? How much information is extracted from the available data? It is the aim of this work to provide such analytical tools.

We turn to the diagnostic methods that have been developed for monitoring statistical multiple-regression analyses. In fact, 4D-Var is a special case of the generalized least-squares (GLS) problem (Talagrand 1997) for weighted regression, thoroughly investigated in the statistical literature.

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The structure of many regression datasets makes effective diagnosis and fitting a delicate matter. In robust (resistant) regression, one specific issue is to provide protection against distortion by anomalous data. In fact, a single unusual observation can heavily distort the results of ordinary (non-robust) least-squares (LS) regression (Hoaglin *et al.* 1982). Unusual or influential data points are not necessarily bad data points; they may contain some of the most useful sample information. For practical data analysis, it helps to judge such effects quantitatively. A convenient diagnostic measures the effect of a (small) change in the observation y_i on the corresponding predicted (estimated) value $\widehat{y}_i$. In LS regression this involves a straightforward calculation: any change in y_i has a proportional impact on $\widehat{y}_i$. The desired information is available in the diagonal of the 'hat matrix' (Velleman and Welsh 1981) which gives the estimated values $\widehat{y}_i$ as a linear combination of the observed values y_i. The term 'hat matrix' was introduced by J. W. Tukey (Tukey 1972) because the matrix maps the observation vector $\mathbf{y}$ into $\widehat{\mathbf{y}}$, but it is also referred to as the 'influence matrix' since its elements indicate the data influence on the regression fit of the data. The matrix elements have also been referred to as the 'leverage' of the data points: in a case of high leverage a unit y value will highly disturb the fit (Hoaglin and Welsh 1978). Concepts related to the influence matrix also provide diagnostics on the change that would occur by leaving out one data point, and the effective information content (degrees of freedom for signal) in the data.

These influence-matrix, $\mathbf{I}_m$, diagnostics are explained in section 2 for ordinary LS regression. In section 3 we derive the corresponding concepts for linear statistical DA schemes. It will be shown that observational influence and background influence complement each other. Thus, for any observation y_i either very large or very small influence could be a sign of inadequacy in the assimilation, and may require further investigation. A practical approximate method that enables calculation of the diagonal elements of the influence matrix for large-dimension variational schemes (such as ECMWF's operational 4D-Var system) is derived in subsection 3(c). In section 4 we present results and selected examples related to data influence diagnostics, including an investigation into the effective information content in several of the main types of observational data. Conclusions are drawn in section 5.

2. Classical statistical definitions of influence matrix and self-sensitivity

The ordinary linear-regression model can be written:

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}, \tag{2.1}$$

where $\mathbf{y}$ is an $m \times 1$ vector for the response variable (predictand); $\mathbf{X}$ is an $m \times q$ matrix of q predictors; $\boldsymbol{\beta}$ is a $q \times 1$ vector of parameters to be estimated (the regression coefficients) and $\boldsymbol{\varepsilon}$ is an $m \times 1$ vector of errors (or fluctuations) with expectation $\mathrm{E}(\boldsymbol{\varepsilon}) = 0$ and covariance $\mathrm{var}(\boldsymbol{\varepsilon}) = \sigma^2\mathbf{I}_m$ (that is, uncorrelated observation errors). In fitting the model (2.1) by LS, the number of observations, m, has to be greater than the number of parameters, q, in order to have a well-posed problem, and $\mathbf{X}$ is assumed to have full rank q.

The LS method provides the solution of the regression equation as $\boldsymbol{\beta} = (\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\mathbf{y}$. The fitted (or estimated) response vector, $\mathbf{y}$, is thus:

$$\widehat{\mathbf{y}} = \mathbf{S}\mathbf{y}, \tag{2.2}$$

where

$$\mathbf{S} = \mathbf{X}(\mathbf{X}^{\mathrm{T}}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}} \tag{2.3}$$

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is the $m \times m$ influence matrix (or hat matrix). It is easily seen that

$$\mathbf{S} = \frac{\partial \widehat{\mathbf{y}}}{\partial \mathbf{y}}, \tag{2.4}$$

and that

$$\begin{aligned} S_{ij} &= \frac{\partial \widehat{y}_j}{\partial y_i} \\ S_{ii} &= \frac{\partial \widehat{y}_i}{\partial y_i}, \end{aligned} \tag{2.5}$$

for the off-diagonal ($i \neq j$) and the diagonal ($i = j$) elements, respectively. Thus, S_{ij} is the rate of change of $\widehat{y}_i$ with respect to y_j variations. The diagonal element S_{ii}, instead, measures the rate of change of the regression estimate $\widehat{y}_i$ with respect to variations in the corresponding observation y_i. For this reason the 'self-sensitivity' (or 'self-influence', or 'leverage') of the ith data point is the ith diagonal element S_{ii}, while an off-diagonal element is a 'cross-sensitivity' diagnostic between two data points.

Hoaglin and Welsh (1978) discuss some properties of the influence matrix. The diagonal elements satisfy

$$0 \leq S_{ii} \leq 1 \quad i = 1, 2, \ldots, m, \tag{2.6}$$

as $\mathbf{S}$ is a symmetric and idempotent projection matrix ($\mathbf{S} = \mathbf{S}^2$). The covariance of the error in the estimate $\widehat{\mathbf{y}}$, and the covariance of the residual, $\mathbf{r} = \mathbf{y} - \widehat{\mathbf{y}}$, are related to $\mathbf{S}$ by:

$$\begin{aligned} \mathrm{var}(\widehat{\mathbf{y}}) &= \sigma^2 \mathbf{S} \\ \mathrm{var}(\mathbf{r}) &= \sigma^2 (I_m - \mathbf{S}). \end{aligned} \tag{2.7}$$

The trace of the influence matrix is:

$$\mathrm{tr}(\mathbf{S}) = \sum_{i=1}^{m} S_{ii} = q = \mathrm{rank}(\mathbf{S}), \tag{2.8}$$

(in fact $\mathbf{S}$ has m eigenvalues equal to 1, and $m - q$ zeros). Thus, the trace is equal to the number of parameters. The trace can be interpreted as the amount of information extracted from the observations or 'degrees of freedom for signal' (Wahba *et al.* 1995). The complementary trace, $\mathrm{tr}(\mathbf{I} - \mathbf{S}) = m - \mathrm{tr}(\mathbf{S})$, on the other hand, is the 'degree of freedom for noise', or simply the degree of freedom (df) of the error variance, widely used for model checking (F test).

A zero self-sensitivity, $S_{ii} = 0$, indicates that the ith observation has had no influence at all in the fit, while $S_{ii} = 1$ indicates that an entire degree of freedom (effectively one parameter) has been devoted to fitting just that data point. The average self-sensitivity value is q/m, and an individual element S_{ii} is considered 'large' if its value is greater than three times the average (Velleman and Welsh 1981). By a symmetrical argument a self-sensitivity value that is less than one-third of the average is considered 'small'.

Furthermore, the change in the estimate that occurs when the ith observation is deleted is:

$$\widehat{y}_i - \widehat{y}_i^{(-i)} = \frac{S_{ii}}{(1 - S_{ii})} r_i, \tag{2.9}$$

where $\widehat{y}_i^{(-i)}$ is the LS estimate of y_i obtained by leaving out the ith observation of the vector $\mathbf{y}$ and the ith row of the matrix $\mathbf{X}$. The method is useful to assess the quality of

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the analysis by using the discarded observation, but impractical for large systems. The formula shows that the impact of deleting $(y_i, \mathbf{x}_i)$ on $\widehat{y}_i$ can be computed by knowing only the residual r_i, and the diagonal element S_{ii}—the nearer the self-sensitivity S_{ii} is to one, the more impact on the estimate $\widehat{y}_i$. A related result concerns the so-called cross-validation (CV) score: that is, the LS objective function obtained when each data point is deleted in turn (Whaba 1990, theorem 4.2.1):

$$\sum_{i=1}^{m} (y_i - \widehat{y}_i^{(-i)})^2 = \sum_{i=1}^{m} \frac{(y_i - \widehat{y}_i)^2}{(1 - S_{ii})^2}. \tag{2.10}$$

This theorem shows that the CV score can be computed by relying on the all-data estimate $\widehat{\mathbf{y}}$ and the self-sensitivities, without actually performing m separate LS regressions on the leaving-out-one samples. Moreover, (2.9) shows how to compute self-sensitivities by the leaving-out-one experiment.

The definitions of influence matrix (2.4) and self-sensitivity (2.5) are rather general, and can be applied also to non-LS and nonparametric statistics. In spline regression, for example, the interpretation remains essentially the same as in ordinary linear regression and most of the results, like the CV theorem above, still apply. In this context, Craven and Wahba (1979) proposed the generalized CV score, replacing S_{ii} in (2.10) by the mean $\mathrm{tr}(\mathbf{S})/q$. For further applications of influence diagnostics beyond usual LS regression (and further references) see Ye (1998) and Shen *et al.* (2002). The notions related to the influence matrix that we have introduced here will be derived in the following section, in the context of a statistical analysis scheme used for DA in numerical weather prediction (NWP).

3. Observational influence and self-sensitivity for a DA scheme

(*a*) *Linear statistical estimation in NWP*

DA systems for NWP provide estimates of the atmospheric state $\mathbf{x}$, by combining meteorological observations $\mathbf{y}$, with prior (or background) information $\mathbf{x}_\mathrm{b}$. A simple Bayesian normal model provides the solution as the posterior expectation for $\mathbf{x}$, given $\mathbf{y}$ and $\mathbf{x}_\mathrm{b}$. The same solution can be achieved from a classical 'frequentist' approach, based on a statistical linear analysis scheme providing the best linear unbiased estimate (Talagrand 1997) of $\mathbf{x}$, given $\mathbf{y}$ and $\mathbf{x}_\mathrm{b}$. The optimal GLS solution to the analysis problem (see Lorenc 1986) can be written:

$$\mathbf{x}_\mathrm{a} = \mathbf{K}\mathbf{y} + (\mathbf{I}_n - \mathbf{K}\mathbf{H})\mathbf{x}_\mathrm{b}. \tag{3.1}$$

The vector $\mathbf{x}_\mathrm{a}$ is the 'analysis'. The gain matrix $\mathbf{K}(n \times p)$ takes into account the respective accuracies of the background vector $\mathbf{x}_\mathrm{b}$ and the observation vector $\mathbf{y}$, as defined by the $n \times n$ covariance matrix $\mathbf{B}$ and the $p \times p$ covariance matrix $\mathbf{R}$, with:

$$\mathbf{K} = (\mathbf{B}^{-1} + \mathbf{H}^\mathrm{T}\mathbf{R}^{-1}\mathbf{H})^{-1}\mathbf{H}^\mathrm{T}\mathbf{R}^{-1}. \tag{3.2}$$

Here, $\mathbf{H}$ is a $p \times n$ matrix interpolating the background fields to the observation locations, and transforming the model variables to observed quantities (e.g. radiative transfer calculations transforming the models temperature, humidity and ozone into brightness temperatures as observed by several satellite instruments). In the 4D-Var context introduced below, $\mathbf{H}$ is defined to also include the propagation in time of the atmospheric state vector to the observation times using a forecast model.

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Substituting (3.2) into (3.1) and projecting the analysis estimate onto the observation space, the estimate becomes:

$$\widehat{\mathbf{y}} = \mathbf{H}\mathbf{x}_\mathrm{a} = \mathbf{H}\mathbf{K}\mathbf{y} + (\mathbf{I}_p - \mathbf{H}\mathbf{K})\mathbf{H}\mathbf{x}_\mathrm{b}. \tag{3.3}$$

It can be seen that the analysis state in observation space ($\mathbf{H}\mathbf{x}_\mathrm{a}$) is defined as a sum of the background (in observation space, $\mathbf{H}\mathbf{x}_\mathrm{b}$) and the observations $\mathbf{y}$, weighted by the $p \times p$ square matrices $\mathbf{I} - \mathbf{H}\mathbf{K}$ and $\mathbf{H}\mathbf{K}$, respectively.

Equation (3.3) is the analogue of (2.1), except for the last term on the right-hand side. In this case, for each unknown component of $\mathbf{H}\mathbf{x}$ there are two data values: a real and a 'pseudo' observation. The additional term in (3.3) includes these pseudo-observations, representing prior knowledge provided by the observation-space background $\mathbf{H}\mathbf{x}_\mathrm{b}$. From (3.3) and (2.4), the analysis sensitivity with respect to the observations is obtained:

$$\mathbf{S} = \frac{\partial \widehat{\mathbf{y}}}{\partial \mathbf{y}} = \mathbf{K}^\mathrm{T}\mathbf{H}^\mathrm{T}. \tag{3.4}$$

Similarly, the analysis sensitivity with respect to the background (in observation space) is given by:

$$\frac{\partial \widehat{\mathbf{y}}}{\partial (\mathbf{H}\mathbf{x}_\mathrm{b})} = \mathbf{I} - \mathbf{K}^\mathrm{T}\mathbf{H}^\mathrm{T} = \mathbf{I}_p - \mathbf{S}. \tag{3.5}$$

We focus here on the expressions (3.4) and (3.5). The influence matrix for the weighted regression DA scheme is actually more complex (see appendix), but it obscures the dichotomy of the sensitivities between data and model in observation space.

The (projected) background influence is complementary to the observation influence. For example, if the self-sensitivity with respect to the ith observation is S_{ii}, the sensitivity with respect the background projected at the same variable, location and time will simply be $1 - S_{ii}$. It also follows that the complementary trace, $\mathrm{tr}(\mathbf{I} - \mathbf{S}) = p - \mathrm{tr}(\mathbf{S})$, is not the *df* for noise but for background, instead; that is, the weight given to prior information, to be compared to the observational weight $\mathrm{tr}(\mathbf{S})$. These are the main differences with respect to standard LS regression. Note that the different observations can have different units, so that the units of the cross-sensitivities are the corresponding unit ratios. Self-sensitivities, however, are pure numbers (no units) as in standard regression. Finally, as long as $\mathbf{R}$ is diagonal, (2.6) is assured, but for more general non-diagonal $\mathbf{R}$ matrices it is easy to find counter examples to that property.

Inserting (3.2) into (3.4), we obtain:

$$\mathbf{S} = \mathbf{R}^{-1}\mathbf{H}(\mathbf{B}^{-1} + \mathbf{H}^\mathrm{T}\mathbf{R}^{-1}\mathbf{H})^{-1}\mathbf{H}^\mathrm{T}. \tag{3.6}$$

As $(\mathbf{B}^{-1} + \mathbf{H}^\mathrm{T}\mathbf{R}^{-1}\mathbf{H})^{-1}$ is equal to the analysis-error covariance matrix $\mathbf{A}$, we can also write $\mathbf{S} = \mathbf{R}^{-1}\mathbf{H}\mathbf{A}\mathbf{H}^\mathrm{T}$.

(*b*) *An idealized case for illustration*

Assume there are two observations, each coincident with a point of the background—that is $\mathbf{H} = \mathbf{I}_2$. Assume the errors of the background at the two locations have correlation $\forall$, that is $\mathbf{B} = \sigma_\mathrm{b}^2 \left(\begin{smallmatrix} 1 & \alpha \\ \alpha & 1 \end{smallmatrix}\right)$, with variance σ_b^2, and that $\mathbf{R} = \sigma_\mathrm{o}^2 \left(\begin{smallmatrix} 1 & 0 \\ 0 & 1 \end{smallmatrix}\right)$. For this simple case $\mathbf{S}$ is obtained with:

$$S_{11} = S_{22} = \frac{r + 1 - \alpha^2}{r^2 + 2r + 1 - \alpha^2}, \tag{3.7}$$

$$S_{12} = S_{21} = \frac{\alpha r}{r^2 + 2r + 1 - \alpha^2}, \tag{3.8}$$

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where $r = \sigma_o^2/\sigma_b^2$. We can see that if the observations are very close (compared to the scale-length of the background-error correlation), i.e. $\alpha \sim 1$, then:

$$S_{11} = S_{22} = S_{12} = S_{21} \simeq \frac{1}{r+2}. \tag{3.9}$$

Furthermore, if $\sigma_b = \sigma_o$, that is $r = 1$, we have three pieces of information with equal accuracy and $S_{11} = S_{22} = 1/3$. The background sensitivity at both locations is $1 - S_{11} = 1 - S_{22} = 2/3$. If the observation is much more accurate than the background ($\sigma_b \gg \sigma_o$), that is $r \sim 0$, then both observations have influence $S_{11} = S_{22} = 1/2$, and the background sensitivities are $1 - S_{11} = 1 - S_{22} = 1/2$.

We now turn to the dependence on the background-error correlation, α, for the case $\sigma_b = \sigma_o$ ($r = 1$). We have:

$$S_{11} = S_{22} = \frac{2-\alpha^2}{4-\alpha^2}, \tag{3.10}$$

$$S_{12} = S_{21} = \frac{\alpha}{4-\alpha^2}. \tag{3.11}$$

If the locations are far apart, such that $\alpha \sim 0$, we obtain $S_{11} = S_{22} = 1/2$; the background sensitivity is also $\frac{1}{2}$ and $S_{12} = S_{21} = 0$. We can conclude that where observations are sparse, S_{ii} and the background sensitivity are determined by their relative accuracies (r), and the off-diagonal terms are small (indicating that surrounding observations have small influence). Conversely, where observations are dense, S_{ii} tends to be small, the background sensitivities tend to be large and the off-diagonal terms are also large.

It is also convenient to summarize the case $\sigma_b = \sigma_o$ ($r = 1$) by showing the projected analysis at location 1:

$$\widehat{y}_1 = \frac{1}{4-\alpha^2}\{(2-\alpha^2)y_1 + 2x_1 - \alpha(x_2 - y_2)\}. \tag{3.12}$$

The estimate $\widehat{y}_1$ depends on y_1, x_1 and an additional term due to the second observation. We see that, with a diagonal $\mathbf{R}$, the observational contribution is generally devalued with respect to the background because a group of correlated background values count more than the single observation ($\alpha \to \pm 1$, $(2-\alpha^2) \to 1$). From the expression above we also see that the contribution from the second observation is increasing with the correlation's absolute value, implying a larger contribution due to the background x_2 and observation y_2 nearby observation y_1.

(*c*) *Approximate calculation of self-sensitivity in a large variational analysis system*

In an optimal variational analysis scheme, the analysis-error covariance matrix, $\mathbf{A}$, is approximately the inverse of the matrix of second derivatives (the Hessian) of the cost function J, i.e. $\mathbf{A} = (\mathbf{J}'')^{-1}$ (Rabier and Courtier 1992). Given the large dimension of the matrices involved, $\mathbf{J}''$ and its inverse cannot be computed explicitly. Following Fisher and Courtier (1995) we use an approximate representation of the Hessian based on a truncated eigenvector expansion with vectors obtained through the Lanczos algorithm. The calculations are performed in terms of a transformed variable Π, $\mathbf{\Pi} = \mathbf{L}^{-1}(\mathbf{x} - \mathbf{x}_b)$, with $\mathbf{L}$ chosen such that $\mathbf{B}^{-1} = \mathbf{L}^T\mathbf{L}$. Thus the transformation $\mathbf{L}$ reduces the covariance of the prior to the identity matrix. In variational assimilation, $\mathbf{L}$ is referred to as the change-of-variable operator (Courtier *et al.* 1998).

$$\boldsymbol{J}''^{-1} \simeq \mathbf{B} - \sum_{i=1}^{M} \frac{1-\lambda_i}{\lambda_i}(\mathbf{L}v_i)(\mathbf{L}v_i)^T. \tag{3.13}$$

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The summation in (3.13) approximates the variance reduction, **B**–**A**, due to the use of observations in the analysis. (λ_i, ν_i) are the eigen-pairs of **A**. In ECMWF's operational DA system, the variances of analysis error are computed according to this method. The variances are inflated to provide estimates of short-term-forecast (background-) error variances to be used as background errors in the next analysis cycle (Fisher 1996). The Hessian eigenvectors are also used to precondition the minimization (Fisher and Andersson 2001). The computed eigenvalues are not used to minimize the cost function but only to estimate the analysis covariance matrix. It is well known, otherwise, that the minimization algorithm is analogous to the conjugate-gradient algorithm. Because the minimum is found within an iterative method, the operational number of iterations is sufficient to find the solution (with required accuracy), but does not provide a sufficient number of eigen-pairs to estimate the analysis-error variances.

The diagonal of the background-error covariance matrix **B** in (3.13) is also computed approximately, using the randomization method proposed by Fisher and Courtier (1995). From a sample of N random vectors $\mathbf{u}_i$ (in the space of the control vector $\boldsymbol{\chi}$), drawn from a population with zero mean and unit Gaussian variance, a low-rank representation of **B** (in terms of the atmospheric state variables **x**) is obtained by:

$$\mathbf{B} = \frac{1}{N}\sum_{i=1}^{N}(\mathbf{L}u_i)(\mathbf{L}u_i)^{\mathrm{T}}. \tag{3.14}$$

This approximate representation of **B** has previously been used by Andersson *et al.* (2000) to diagnose background errors in terms of observable quantities, i.e. $\mathbf{HBH}^{\mathrm{T}}$.

Inserting (3.13) and (3.14) into (3.6) we arrive at an approximate method for calculating **S**, that is practical for a large-dimension variational assimilation (both 3D- and 4D-Var):

$$\mathbf{S} = \mathbf{R}^{-1}\mathbf{H}\left\{\frac{1}{N}\sum_{i=1}^{N}(\mathbf{L}u_i)(\mathbf{L}u_i)^{\mathrm{T}} + \sum_{i=1}^{M}\frac{1-\lambda_i}{\lambda_i}(\mathbf{L}v_i)(\mathbf{L}v_i)^{\mathrm{T}}\right\}\mathbf{H}^{\mathrm{T}}. \tag{3.15}$$

Only the diagonal elements of **S** are computed and stored—that is, the analysis sensitivities with respect to the observations, or self-sensitivities S_{ii}. The cross-sensitivity S_{ij} for $i \neq j$, that represents the influence of the jth observation to the analysis at the ith location, is not computed. Note that the approximation of the first term is unbiased, whereas the second term is truncated such that variances are underestimated. For small M the approximate S_{ii} will tend to be overestimates. For the extreme case $M = 0$, (3.15) gives $\mathbf{S} = \mathbf{R}^{-1}\mathbf{HBH}^{\mathrm{T}}$ which, in particular, can have diagonal elements larger than one if elements of $\mathbf{HBH}^{\mathrm{T}}$ are larger than the corresponding elements of **R**. We investigate the effects of the approximations on the calculated S_{ii} values in subsection 4(d).

Note that (2.9) can be applied to quantify how much the analysis at a given observation location would be changed by deleting the observation itself. The change depends only on the self-sensitivity and the residual value at that location. Note also that the tr(**S**) provides estimates of the information content of the data with respect to the background, and is equal to the degrees of freedom for signal as studied by e.g. Purser and Huang (1993), Rabier *et al.* (2002) and Fourrié and Thépaut (2003) in the context of remote-sensing retrieval applications. In particular, Rabier *et al.* (2002) use a 'data resolution matrix', which is basically the same as the influence matrix. Recently Fisher (2003) computed an estimate of the global tr(**S**) by using the Bay *et al.* (1996) method, without explicitly computing the individual elements S_{ii}. Comparison with Fisher's (2003) estimate has provided validation of our method, in terms of the global trace.

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TABLE 1. DATA COUNTS BY OBSERVATION TYPE 1200 UTC 13 FEBRUARY 2003

Type of data	Description	No. of observations
SYNOP	Surface observations from land stations and ships, measuring: p_s, T, rh, u and v	36 889
QuikSCAT	Satellite microwave scatterometer, measuring: u and v at the ocean surface	114 220
DRIBU	Drifting buoy measuring: p_s, T, rh, u and v	3 419
SATOB	Satellite cloud drift winds	102 090
TEMP	Radiosonde from land stations and ships measuring: p_s, and tropospheric and stratospheric values of T, rh, u and v	61 205
AMSUA	Satellite microwave sounder radiances	641 274
PILOT	Radiosonde and wind profiler measurements of upper-air u and v	45 060
HIRS	Satellite infrared radiances	163 134
AIREP	Aircraft measurements of T, u and v, at a single upper level	119 545
SSMI	Satellite microwave imager radiances	116 739
PAOB	Bogus observations of p_s	219
GOES	Geostationary satellite infrared sounder radiances	35 302
OZONE	Satellite ozone retrieval	11 058
METEOSAT	Geostationary satellite infrared sounder radiances	39 623
Grand total		1 489 777

Here p_s is surface pressure, T is temperature, rh is relative humidity, u and v are components of wind velocity.

4. RESULTS

The diagonal elements of the influence matrix have been computed for the operational 4D-Var assimilation system. The calculations in (3.15) have been carried out on 60 model levels at T95 spectral truncation $n = 2\,802\,912$, which is the resolution used for estimation of analysis and background-error variances in the operational system. The observation departures ($\mathbf{y} - \mathbf{H}\mathbf{x}_b$) were calculated by comparing the observations with a 12 h forecast integration at T319 resolution. The experiment time is 12 UTC 13 February 2003 using observations between 03 and 15 UTC. The number of Hessian vectors used here is $M = 753$, and the number of random $\mathbf{B}$ vectors is $N = 500$ ($M = 40$ and $N = 50$ in operations). The counts of assimilated observations for each main observation type are given in Table 1. together with brief details. The global total number of observations in this case is $p = 1\,489\,777$. A large proportion of the data used is provided by satellite systems (Thépaut and Andersson 2003); the remainder are from surface-based observing systems (see WMO 1996).

(*a*) *Self-sensitivity examples*

Self-sensitivities for SYNOP surface pressure observations are shown in Fig. 1. Each box indicates the observation influence at the observation location. Data points with influence greater than one are investigated in subsection 4(d). Low-influence data points have large background influence (see (3.4) and (3.5)), which is the case in data-rich areas such as North America and Europe (observation influence $\sim$0.2). In data-sparse areas individual observations have larger influence: in the Polar regions, where there are only few isolated observations, $S_{ii} \sim 1$ and the background has small influence on the analysis.

In dynamically active areas (Fig. 1: North Atlantic), several fairly isolated observations have a large influence on the analysis. This is due to the evolution of the background-error covariance matrix as propagated by the forecast model in 4D-Var (Thépaut *et al.* 1993, 1996). As a result, the DA scheme can fit these observations more closely.

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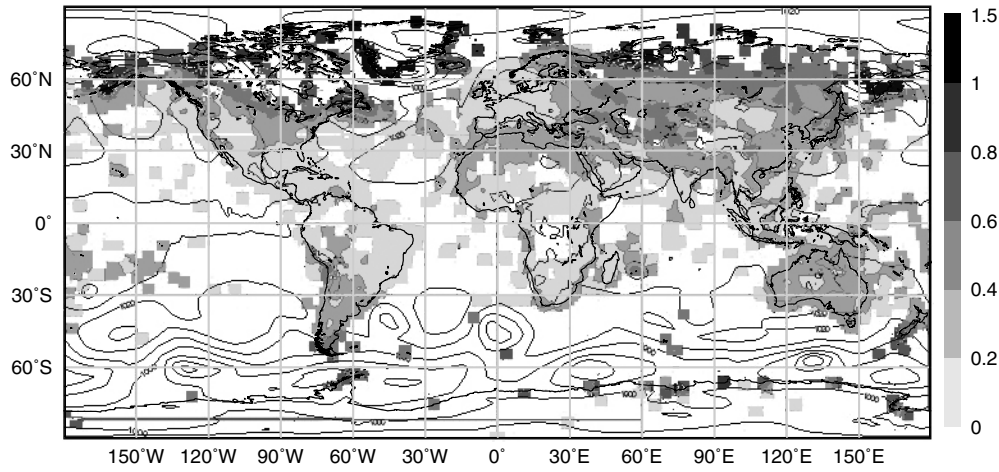


Figure 1. Observation influence of surface pressure SYNOP data valid for 12 UTC 13 February 2003.

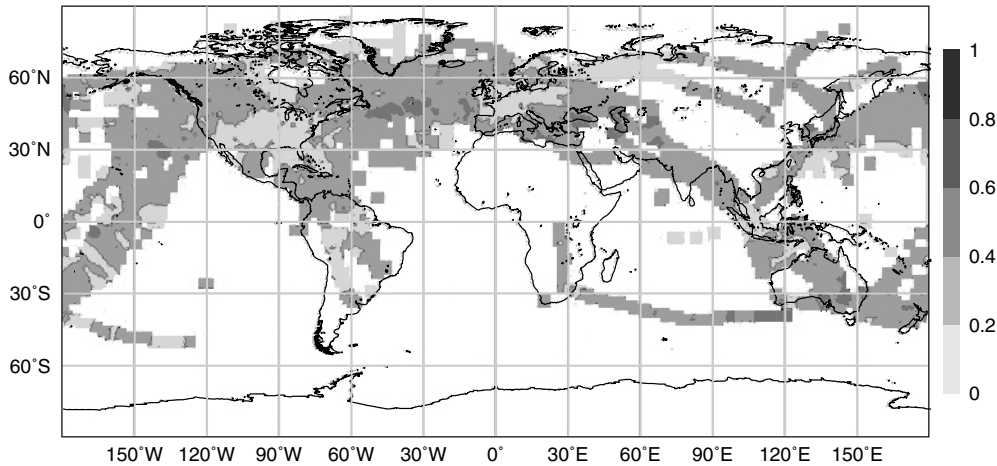


Figure 2. Observation influence of wind u component from AIREP data between 200 and 300 hPa. Analysis experiment valid 12 UTC 13 February 2003.

Similar features can be seen in Fig. 2, which shows the influence of u component wind observations for aircraft (AIREP) data between 300 and 200 hPa. Isolated flight tracks over Atlantic and Pacific oceans show larger influences than measurements over data-dense areas over America and Europe.

(*b*) *Trace diagnostic*

We define the global average influence as the globally averaged observation influence. It is given by

$$GAI = \frac{\mathrm{tr}(\mathbf{S})}{p}, \tag{4.1}$$

where p is the total number of observations. In our experiment we found that $GAI = 0.15$. Consequently, the average background global influence to the analysis at observation points is equal to 0.85 (see (3.5)).

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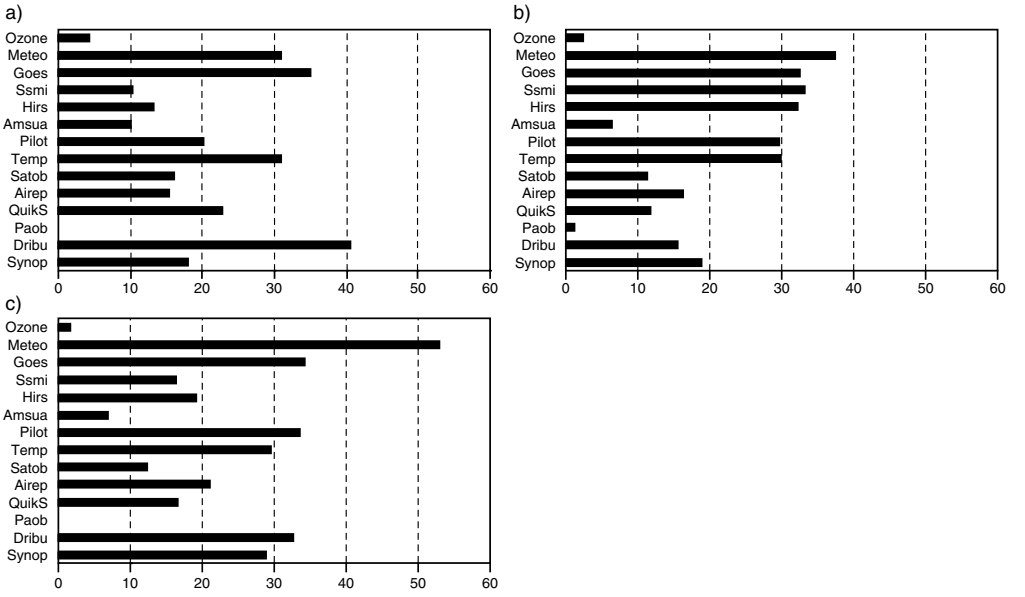


Figure 3. Average self-sensitivity (%) for each of the main observation types (see Table 1): (a) for the northern hemisphere extratropics partial influence $PAI = 15\%$, (b) for the tropics $PAI = 17.5\%$, and (c) for the southern hemisphere extratropics $PAI = 12\%$.

Another index of interest is the partial influence for any selected subset of data given by:

$$PAI = \frac{\sum_{i \in I} S_{ii}}{p_I}, \tag{4.2}$$

where p_I is the number of data in subset I. The subset I can, for example, represent a specific observation type, a specific vertical or horizontal domain, or a particular meteorological variable. In Fig. 3 the *PAI* for the different observation types is plotted for three different geographical areas: the northern hemisphere extratropics ($PAI = 15\%$), the Tropics ($PAI = 17.5\%$) and the southern hemisphere extra tropics ($PAI = 12\%$). It is clear that METEOSAT and GOES radiances have a relatively large average influence ($PAI = 35\%$) in the Tropics and in the southern hemisphere. In the northern hemisphere, DRIBU (drifting buoy) observations have $PAI = 40\%$, mainly from surface pressure measurements. On the other hand, ozone observations have very small average influence on the analysis. Also AMSU-A radiance data have a low average influence $PAI < 10\%$. Similar diagnostics can be obtained for any other data subset of interest. The slightly higher influence in the northern hemisphere compared to the southern hemisphere, is explained by the fact that the influence is higher for isolated observations, and observations in very dynamically active areas especially in data-sparse regions. The last factor is due to the inflation of background-error variances implied by 4D-Var.

As we found relatively high influence from satellite measurements that have strong dependence on atmospheric humidity (i.e. GOES, METEOSAT, SSMI and HIRS), we show in Fig. 4 a scatter plot for all types of humidity-related observations (every type is a subset I). In Fig. 4(a) normalized sub-traces of tr($\mathbf{S}$) are plotted versus p_I/p for the tropical area. The sub-traces represent the influence of relative humidity (rh) from SYNOP and DRIBU, specific humidity (q) from TEMP and radiances from METEOSAT (two satellites), GOES (one satellite), SSMI (three satellites, seven

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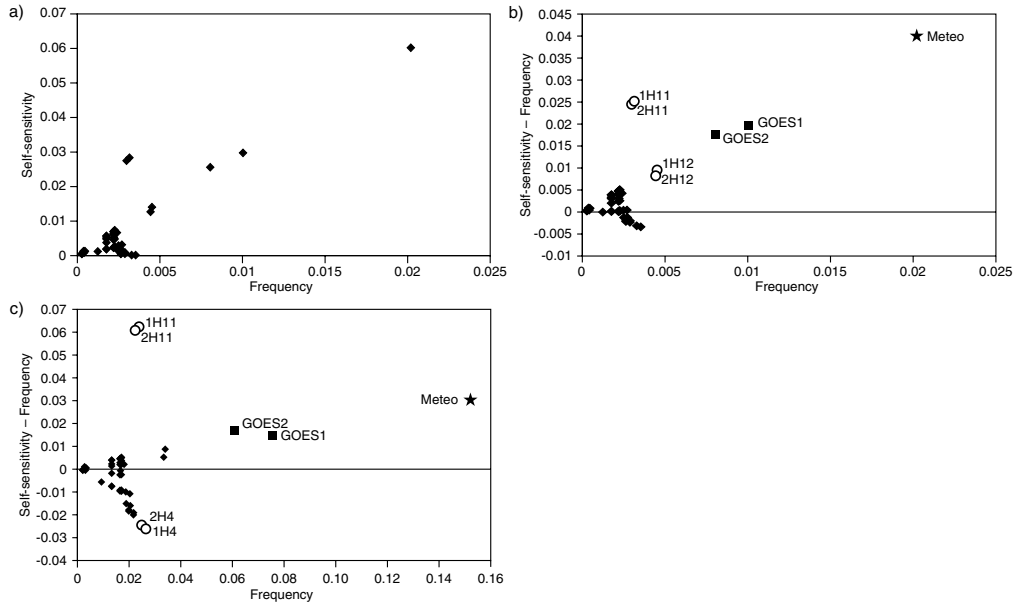


Figure 4. (a) Normalized sub-traces of Tr(**S**) plotted versus p_I/p for the tropical area and for relative humidity from SYNOP and DRIBU, specific humidity from TEMP, and radiances from METEOSAT, GOES, SSMI and HIRS; (b) residuals (self-sensitivities minus frequency) with respect to the bisect line plotted versus p_I/p; (c) as (a) but values are normalized with respect to partial amount. See text and Table 1 for details.

channels) and HIRS (two satellites, eight channels). The sum of all subsets I defines the subset F. On the x axis, the relative number of observations per sub-trace is normalized with respect to the total observation number (frequency $= p_I/p$).

In Fig. 4(b), the residuals ($\sum_{i\in I} S_{ii}/\mathrm{tr}(S) - p_I/p$) with respect to the bisect regression line are plotted versus p_I/p (as before); this highlights humidity data types with particularly high or low self-sensitivity. We find that the radiances from the HIRS (channels 11 and 12), GOES and METEOSAT satellite data types have a significantly higher (p-value $= 0$) self-sensitivity than all the other assimilated data. Also, these observation types are significantly (p-value $= 8\%$) more influential than the other humidity-related observations. In fact, normalizing the residuals with respect to partial amounts ($\sum_{i\in\mathrm{HIRS}} S_{ii}/\sum_{i\in F} S_{ii} + \sum_{i\in\mathrm{SSMI}} S_{ii}/\sum_{i\in F} S_{ii} + \cdots$ and for the x axis $p_{\mathrm{HIRS}}/p_F + p_{\mathrm{SSMI}}/p_F + \cdots$) (see Fig. 4(c)) shows the high influence of HIRS channel 11, in particular. Further investigations of this and similar diagnostics should be performed to find out if this is an indication that aspects of the background-error or observation-error covariances need to be adjusted, or if what we see is the correct behaviour given the relative accuracies of the background and the observation error for the various data types.

(*c*) *Information content*

In section 2 we showed that tr(**S**) can be interpreted as a measure of the amount of information extracted from the observations. In fact, in non-parametric statistics, tr(**S**) measures the 'equivalent number of parameters' or 'degrees of freedom for signal'. Having obtained values of all the diagonal elements of **S** (using (3.15)) we can now obtain reliable estimates of the information content in any subset of the observational data. In Fig. 5 we illustrate this with one example; it must be noted that this theoretical

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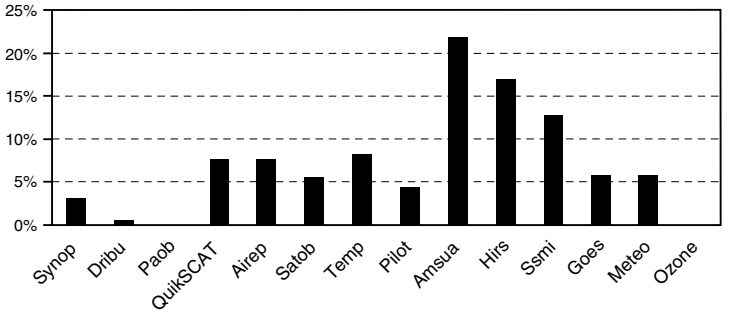


Figure 5. Information content, i.e. degree of freedom for signal, for the main data types in the assimilation. See Table 1 for details of data types.

measure of information content does not necessarily translate to value in terms of forecast impact. Figure 5 shows the information content for all main observation types. We see that AMSU-A radiances are the most informative data type, providing 22% of the total observational information. HIRS follows with 17% and SSMI with 13%. The information content of AIREP, QuikSCAT, TEMP, GOES and METEOSAT is 6–8% each, while SYNOP, SATOB and PILOT are each less than 5%. A large part of the AMSU-A information is with respect to stratospheric temperature. Most of the SSMI and HIRS information is with respect to humidity. DRIBU and OZONE information content is small: ozone observations have a very small average influence (Fig. 3) and dense data coverage, while DRIBU observations have large mean influence per observation but much lower data counts (Fig. 3). The ozone data are important for the ozone assimilation in spite of their low information content per analysis cycle. Ozone is generally a long-lived species, which allows observational information to be advected accurately by the model over periods of several days. The importance of AMSU-A correlates well with recent data-impact studies by (Bouttier and Kelly 2001; Kelly 2003, personal communication).

Our results are in good qualitative agreement with those obtained by Fisher (2003) using the algorithm of Bay *et al.* (1996). The advantage of the influence-matrix approach is that it allows the computation of information content for any subset of data by straightforwardly summing up the elements corresponding to each subset. The methods based on a global trace estimate, however, require that the analysis scheme is re-run once for every data subdivision (Fisher 2003).

(*d*) *High-influence data*

From (2.6) it is known that self-sensitivities cannot be larger than one, but we have seen from Fig. 1 that some S_{ii} are greater than one. In this paragraph the nature of the upper-bound values is explained.

(i) *Self-sensitivity calculation approximation.* In Fig. 6(a) the self-sensitivities for HIRS channel 11 are shown; values greater than one are shown in black. Here only $M = 40$ Hessian eigenvectors and $N = 50$ random $\mathbf{B}$ vectors have been used to compute the self-sensitivities. In this worst case, 67% of the self-sensitivities are greater than one, in contrast with (2.6) which states that the diagonal elements of S are bounded to the interval between zero and one. The occurrence of $S_{ii} > 1$ is partly due to the approximate nature of our method of calculating self-sensitivity. Approximations in both of the two terms of (3.15) contribute to the problem. In the second term the number of

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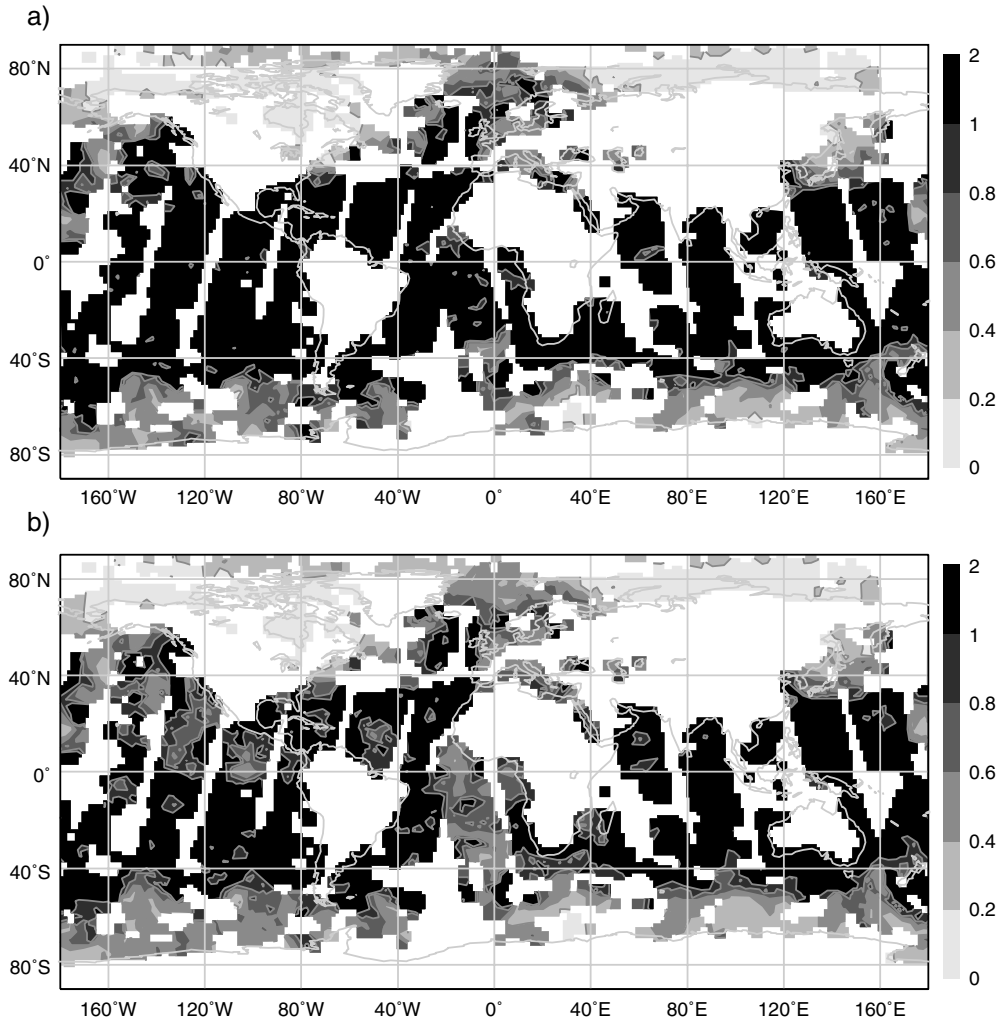


Figure 6. Influence of radiances from HIRS (see Table 1) channel 11 using M Hessian eigenvectors: (a) $M = 40$, (b) $M = 753$.

Hessian eigenvectors is truncated to M; the term is therefore underestimated, and S_{ii} will tend to be overestimated. The degree of overestimation depends on the structure of the covariance reduction matrix $\mathbf{B} - \mathbf{A}$.

For an analysis in which observations lead to strongly localized covariance reduction (such as the humidity analysis with its short covariance length scales ~180 km, and large observational impacts) a large M is required to approximate $\mathbf{B} - \mathbf{A}$ accurately. We repeated the calculations for several values of $M > 40$ to an upper limit of $M = 735$ that, given the available computer resources, could reasonably be afforded. In Fig. 6(b) we show the result corresponding to Fig. 6(a) but for $M = 753$ Hessian vectors. The number of $S_{ii} > 1$ for this HIRS channel is now reduced by 30%. This approximate computation mostly affects the self-sensitivities close to the upper bound leaving the self-sensitivities <0.7 almost unaffected.

HIRS channel 11 is the dataset for which the problem is most acute. In fact, even for $M = 40$ self-sensitivities greater than one only appear for a very small proportion

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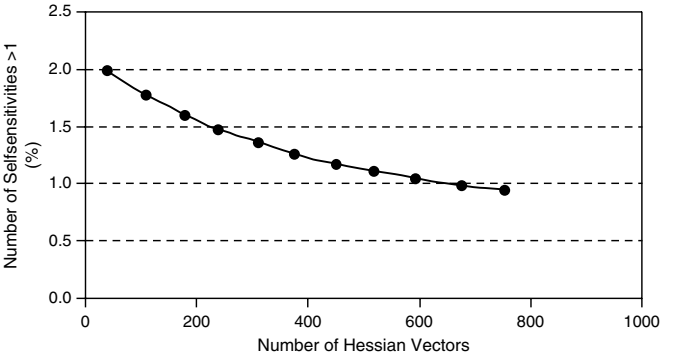


Figure 7. Proportion of self-sensitivity values greater than one (%) versus the number of Hessian vectors used to compute the variances of $\mathbf{B} - \mathbf{A}$, using the approximate method described in (3.15).

(2%) of the global set of observations. In Fig. 7, the proportion of observations for which $S_{ii} > 1$ is plotted against M. The plot shows a gradual decrease of $S_{ii} > 1$ as M increases, as expected. The curve seems to approach 10 000 $S_{ii} > 1$ (0.7% in the plot) for M somewhere between 1000 and 2000. However, increasing the number of Hessian vectors slightly increases the number of self-sensitivities less than zero (by 0.5%). This problem can be explained by looking at the approximations introduced through the first term of (3.15). The truncation N of the first term determines the randomization sample size: larger N leads to smaller noise. The noise is unbiased—that is, the term is neither overestimated nor underestimated on average. The randomization noise in the diagonal elements is in the order 10% with $N = 50$ (Andersson *et al.* 2000). With $N = 500$, values of $S_{ii} < 0$ have all disappeared.

(ii) *Ill-conditioning problem.* A set of linear equations is said to be ill-conditioned if small variations have a large effect on the exact solution $\widehat{\mathbf{y}}$. For example, if $\mathbf{X}$ (or its analogue in 4D-Var) is nearly singular, small changes in its element can cause large variations in DA, and serious effects on the stability and accuracy of the solution. A measure of ill-conditioning is the condition number, K, defined as the ratio of the largest and the smallest non-zero singular value of $\mathbf{X}$, that is, the square roots of the eigenvalue of $\mathbf{X}\mathbf{X}^{\mathrm{T}}$. Regression models are notoriously ill-conditioned when the degree is higher than five.

It is not uncommon that variates (regressors) can be in linear relationship with a set of variates (collinearity). This means that $\mathbf{X}^{\mathrm{T}}\mathbf{X}$ will be near singularity, the smallest eigenvalue will be small and K will be large. In 4D-Var K is typically 10^4 to 10^5. Cases of poor convergence of the 4D-Var solution algorithm have been studied (Andersson *et al.* 2000). It was found that when the ratio $\sigma_{\mathrm{b}}/\sigma_{\mathrm{o}}$ between specified background and observation errors is large, or the observation density is large, the 4D-var problem can be ill-conditioned and poor convergence in the minimization can result.

We have found that self-sensitivities exceeding the upper bound ($S_{ii} > 1$) can provide indications of data that contribute to poor conditioning of the analysis, because of large $\sigma_{\mathrm{b}}/\sigma_o$ ratio. To illustrate this, self-sensitivities computed in a single analysis experiment have been compared to self-sensitivities computed during a modified 4-day assimilation cycle. In Fig. 8(a) self-sensitivities for DRIBU surface pressure observation over the North Atlantic are shown together with the mean-sea-level pressure field. Higher self-sensitivities (>0.6) are close to a dynamically active area (a developing low). Figure 8(b) shows that the ratio $\sigma_{\mathrm{b}}/\sigma_{\mathrm{o}}$ for the same observations is close to one.

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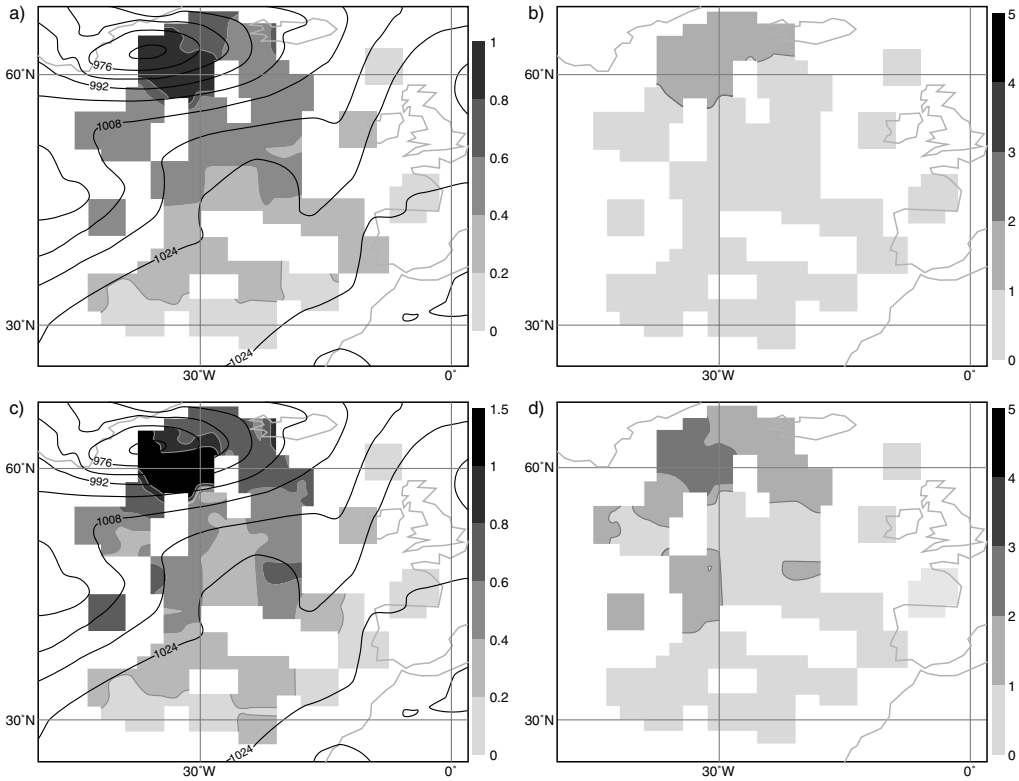


Figure 8. (a) DRIBU (see Table 1) surface pressure influence valid 12 UTC 13 February 2003; (b) ratio between background and observation standard deviations for the experiment shown in (a). (c) DRIBU surface pressure influence from a flow-dependent background standard deviation experiment cycled for 4 days and valid for 12 UTC 13 February 2003; (d) as (b) but for the experiment shown in (c).

The patterns in both Figs. 8(a) and (b) are dependent on the background-error variance specification, which in ECMWF practice is modified to account for error growth in the prediction over the time interval between analyses (12 h). Operationally (and in Figs. 8(a) and (b)), the **B** matrix variances are inflated exponentially in time (Fisher 1996), and the new variances are used in the next analysis cycle. In Figs. 8(c) and (d) the error inflation has instead been performed in a flow-dependent manner using the tangent linear forecast model (Andersson and Fisher 1999). The influence pattern for DRIBU surface pressure is now computed after having cycled the 4D-Var system for four days (the experiment started at 12 UTC 9 February 2003). The results shown in Figs. 8(c) and (d) are valid at the same time as Fig. 8(a) (12 UTC 13 February 2003). In the region of low pressure, self-sensitivities are now larger with some exceeding one (Fig. 8(c)). Also the ratio σ_b/σ_o (Fig. 8(d)) has increased, in some locations by a factor of three. Variance inflation and propagation (applied to the background standard deviation) can thus generate collinearity, and potentially ill-condition the estimation problem. We know that data points with self-sensitivities greater than one are due to the approximate nature of our computation but, nevertheless, when they occur they can be valuable by drawing attention to those data points that may be contributing towards ill-conditioning of 4D-Var.

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5. Conclusions

The influence matrix is a well-known concept in multivariate linear regression, where it is used to identify influential data and to predict the impact on the estimates of removing individual data from the regression. In this paper we have derived the influence matrix in the context of linear statistical analysis schemes, as used for the assimilation of meteorological observations in NWP (Lorenc 1986). In particular we derive and implement an approximate method to compute the diagonal elements of the influence matrix (the self-sensitivities) in ECMWF's operational DA system (4D-Var). Our approach is necessarily approximate due to the large dimension of the estimation problem at hand: the number of estimated parameters is of the order 10^6 (8×10^6 in operational practice, and 3×10^6 in the calculations presented here), and the number of observational data is around 1.5×10^6.

The self-sensitivity provides a quantitative measure of the observation influence in the analysis. In robust regression it is expected that data have similar self-sensitivity (sometimes called leverage)—that is, they exert similar influence in estimating the regression line. Disproportionate data influence on the regression estimate can have different reasons: first, there is the inevitable occurrence of incorrect data; second, influential data points may be correct extreme observations. However, even if such data often contain valuable information, it is constructive to determine to what extent the estimate depends on them. Moreover, diagnostics may reveal other patterns, e.g. that the estimates are based primarily on a specific sub-set rather than on the majority of the data. We provided an illustration with respect to humidity-related observations in 4D-Var showing particularly large influence of HIRS channel 11 (Fig. 6).

In the context of 4D-Var there are many components that together determine the influence given to any one particular observation. First there is the specified observation-error covariance **R**, which is usually well known and obtained simply from tabulated values. Second, there is the background-error covariance **B**, which is specified in terms of transformed variables that are most suitable to describe a large proportion of the actual background-error covariance. The implied covariance in terms of the observable quantities is not immediately available for inspection, but it determines the analysis weight given to the data. Third, the dynamics and the physics of the forecast model propagate the covariance in time, and modify it according to local error growth in the prediction. The influence is further modulated by data density. We showed examples for surface pressure and aircraft wind observations indicating that low-influence data points occur in data-rich areas while high-influence data points are in data-sparse regions or in dynamically active areas. Background-error correlations also play an important role. In fact, very high correlations drastically lessen the observation influence (it is halved in the idealized example presented in subsection 3(b)) in favour of background influence, and amplify the influence of the surrounding observations.

In this study the global observation influence per assimilation cycle has been found to be 15%, and consequently the background influence is 85%. Thus, on average the observation influence is low compared to the influence of the background (the prior). However, it must be taken into account that the background contains observation information from the previous analysis cycles. The theoretical information content (the degrees of freedom for signal) for each of the main observation types was also calculated. It was found that AMSU-A radiance data provide the most information to the analysis (22%), followed by HIRS (17%), SSMI (13%), and then AIREP, QuikSCAT, TEMP, Goes and Meteosat each providing 6–8 %. It must be stressed that this ranking is not an indication of relative importance of the observing systems for forecast accuracy. The results compare well with independent estimates provided by Fisher (2003).

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If the influence matrix was computed without approximation, then all the self-sensitivities would be bounded in the interval zero to one. With the approximate method used here, out-of-bound self-sensitivities occur if the Hessian representation based on an eigenvector expansion is truncated, especially when few eigenvectors are used. However, it has been shown that this problem affects only a small percentage of the self-sensitivities computed in this study, and in particular those that are closer to one. Remaining values greater than one can be due to large background-error to observation-error ratio, which is one factor that is known to contribute towards ill-conditioning and poor convergence of the 4D-Var algorithm.

Self-sensitivities provide an objective diagnostic on the performance of the assimilation system. They could be used in observation quality-control to protect against distortion by anomalous data (however this aspect has not been explored within the current study). In fact, the leaving-out-one observation scheme, which is not practical for a large system dimension, uses the discarded observation to assess the quality of the analysis. It has been shown (2.9) that self-sensitivities provide a similar diagnostic without performing separate LS regressions. Self-sensitivities also provide indications regarding model and observation error, specification and tuning. Incorrect specifications can be identified, interpreted and better understood through observation-influence diagnostics, partitioned by e.g. observation types, variable, levels, and regions.

In the near future more satellite data will be used and they are likely be 'thinned'. Thinning has to be undertaken either to reduce the observation-error spatial correlation (Bormann *et al.* 2003) or to reduce the computational cost of the assimilation. The observation influence provides an objective way of selecting observations dependent on their local influence on the analysis estimate, to be used in conjunction with forecast-impact assessments. It would be interesting to see if a small sub-group of very influentialial data (i.e. satellite observations) have the same impact in the forecast as the full amount of data. If this is the case, a dynamical thinning can be devised that selects, for every assimilation cycle, the most influential partition of a particular remote-sensing instrument measurements, from information based on the previous cycle. Clearly, it can be assumed that components of the observing network remain constant and background-error variances remain almost unchanged for close assimilation cycles.

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APPENDIX

Influence-matrix calculation in the weighted regression data assimilation scheme

Under the 'frequentist' approach, the regression equations for observation:

$$\mathbf{y} = \mathbf{H}\boldsymbol{\theta} + \boldsymbol{\varepsilon}_{\mathrm{o}},$$

and for background:

$$\mathbf{x}_{\mathrm{b}} = \boldsymbol{\theta} + \boldsymbol{\varepsilon}_{\mathrm{b}},$$

are assumed to have uncorrelated error vectors $\boldsymbol{\varepsilon}_{\mathrm{o}}$ and $\boldsymbol{\varepsilon}_{\mathrm{b}}$, zero vector means and variance matrices $\mathbf{R}$ and $\mathbf{B}$, respectively. The $\boldsymbol{\theta}$ parameter is the unknown system state ($\mathbf{x}$) of

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dimension n. These regression equations are summarized as a weighted regression:

$$\mathbf{z} = \mathbf{X}\boldsymbol{\theta} + \boldsymbol{\varepsilon},$$

where $\mathbf{z} = [\mathbf{y}^{\mathrm{T}}\mathbf{x}_{\mathrm{b}}^{\mathrm{T}}]^{\mathrm{T}}$ is $(p+n)\times 1$; $\mathbf{X} = [\mathbf{H}^{\mathrm{T}}\mathbf{I}_n]^{\mathrm{T}}$ is $(p+n)\times n$ and $\boldsymbol{\varepsilon} = [\boldsymbol{\varepsilon}_{\mathrm{o}}\boldsymbol{\varepsilon}_{\mathrm{b}}]^{\mathrm{T}}$ is $(m+n)\times 1$ with zero mean and variances matrix:

$$\boldsymbol{\Omega} = \begin{pmatrix} \mathbf{R} & 0 \\ 0 & \mathbf{B} \end{pmatrix}.$$

The generalized LS solution for $\boldsymbol{\theta}$ is BLUE and is given by:

$$\widehat{\boldsymbol{\theta}} = (\mathbf{X}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{z}, \tag{A.1}$$

see Talagrand (1997). After some algebra this equation equals (3.1). Thus:

$$\mathbf{z} = \mathbf{X}\widehat{\boldsymbol{\theta}} = [\mathbf{H}^{\mathrm{T}}\mathbf{x}_{\mathrm{a}}^{\mathrm{T}}\mathbf{x}_{\mathrm{a}}^{\mathrm{T}}]^{\mathrm{T}} = \mathbf{X}(\mathbf{X}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{X})^{-1}\mathbf{X}^{\mathrm{T}}\boldsymbol{\Omega}^{-1}\mathbf{z},$$

and by (2.5) the influence matrix becomes:

$$\mathbf{S}_{zz} = \frac{\partial\widehat{\mathbf{z}}}{\partial\mathbf{z}} = \frac{\partial\mathbf{X}\widehat{\boldsymbol{\theta}}}{\partial\mathbf{z}} = \begin{pmatrix} \mathbf{S}_{yy} & \mathbf{S}_{y\mathrm{b}} \\ \mathbf{S}_{\mathrm{b}y} & \mathbf{S}_{\mathrm{bb}} \end{pmatrix} = \begin{pmatrix} \mathbf{R}^{-1}\mathbf{H}\mathbf{A}\mathbf{H}^{\mathrm{T}} & \mathbf{R}^{-1}\mathbf{H}\mathbf{A} \\ \mathbf{B}^{-1}\mathbf{A}\mathbf{H}^{\mathrm{T}} & \mathbf{B}^{-1}\mathbf{A} \end{pmatrix},$$

where

$$\mathbf{S}_{yy} = \frac{\partial\mathbf{H}\mathbf{x}_{\mathrm{a}}}{\partial\mathbf{y}}; \quad \mathbf{S}_{y\mathrm{b}} = \frac{\partial\mathbf{x}_{\mathrm{a}}}{\partial\mathbf{y}}; \quad \mathbf{S}_{\mathrm{b}y} = \frac{\partial\mathbf{H}\mathbf{x}_{\mathrm{a}}}{\partial\mathbf{x}_{\mathrm{b}}}; \quad \mathbf{S}_{\mathrm{bb}} = \frac{\partial\mathbf{x}_{\mathrm{a}}}{\partial\mathbf{x}_{\mathrm{b}}}.$$

Note that $\mathbf{S}_{yy} = \mathbf{S}$ as defined in (3.4).

Generalized LS regression is different from ordinary LS because the influence matrix is not symmetric anymore. For idempotence, using (A.1) it easy to show that $\mathbf{S}_{zz}\mathbf{S}_{zz} = \mathbf{S}_{zz}$. Finally:

$$\mathbf{S}_{\mathrm{bb}} = \mathbf{B}^{-1}\mathbf{A} = \mathbf{I}_n - \mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\mathbf{A},$$

hence:

$$\mathrm{tr}(\mathbf{S}_{\mathrm{bb}}) = n - \mathrm{tr}(\mathbf{H}^{\mathrm{T}}\mathbf{R}^{-1}\mathbf{H}\mathbf{A}) = n - \mathrm{tr}(\mathbf{S}_{yy}).$$

It follows that:

$$\mathrm{tr}(\mathbf{S}_{zz}) = \mathrm{tr}(\mathbf{S}_{yy}) + \mathrm{tr}(\mathbf{S}_{\mathrm{bb}}) = n.$$

The trace of the influence matrix is still equal to the parameter's dimension.

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